

2024 AMC 10B Problem 12 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 12

**Problem:**

A group of 100 students from different countries meet at a mathematics competition. Each student speaks the same number of languages, and, for every pair of students A and B, student A speaks some language that student B does not speak, and student B speaks some language that student A does not speak. What is the least possible total number of languages spoken by all the students?

Answer Choices:

- A. 9
- B. 10
- C. 12
- D. 51
- E. 100

Solution:

Suppose the languages spoken are labeled $L_1, L_2, L_3, \dots, L_n$. Note that the collection of all subsets of $\{L_1, L_2, L_3, \dots, L_n\}$ of size r will satisfy the conditions in the problem for any r in the range $1 \leq r < n$. For a given n , there are $\binom{n}{r}$ subsets, and $\binom{n}{r}$ is maximized when $r = \lfloor \frac{n}{2} \rfloor$ or $r = \lceil \frac{n}{2} \rceil$.

If $n = 8$ and $r = 4$, the number of distinct subsets is $\binom{8}{4} = 70 < 100$, so $n > 8$. But $\binom{9}{4} = \binom{9}{5} = 126 > 100$, so $n = \text{(A)}\boxed{9}$ is the least possible total number of languages spoken by all the students.

Note: The restriction on having the students speak the same number of languages is not needed. A collection of subsets of a given set with n elements satisfying the condition that no member of the collection is a subset of another is called an antichain. Sperner's Theorem asserts that the largest antichain occurs when the subsets are all of size r , where $r = \lfloor \frac{n}{2} \rfloor$ or $\lceil \frac{n}{2} \rceil$.

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